Malnad College Of Engineering

Under the auspices of M.T.E.S ®

(An Autonomous Institution Affiliated to VTU, Belgaum) P.B. No. 21, Hassan 573 202, Karnataka

Block-chain Technology Micro project (23IS664)

TOPIC :- "Decentralized Product Verification Using EVM and Cryptographic Hashing "

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2025-26

ABSTRACT

The proliferation of counterfeit goods in global supply chains poses significant economic risks and safety concerns for both manufacturers and consumers. Traditional centralized verification systems often suffer from single points of failure, lack of transparency, and susceptibility to data tampering. This paper proposes a decentralized product verification framework leveraging the Ethereum Virtual Machine (EVM) and cryptographic hashing to ensure product authenticity and end-to-end traceability.

By assigning a unique, immutable digital identity to each physical product—generated via high-security hashing algorithms (such as SHA-256) and stored as a Non-Fungible Token (NFT) or a state entry on the blockchain—the system creates a verifiable "digital twin." Smart contracts deployed on the EVM govern the lifecycle of the product, automating ownership transfers and preventing the double-spending of physical assets.

Preliminary analysis indicates that the integration of EVM-based logic significantly reduces the administrative overhead of verification while providing a robust defense against counterfeit injection, offering a scalable solution for high-value industries like pharmaceuticals, luxury goods, and electronics.

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